Chapter2

Machine Intelligence and Cyber Security in Computing

Define machine intelligence

Machine intelligence (MI) refers to the ability of machines or computers to perform tasks that typically require human intelligence, such as

1. Learning
2. Problem solving
3. Reasoning
4. Perception
5. Language understanding

Machine intelligence is achieved through

1. Machine learning (ML)
2. Deep learning
3. Natural language processing
4. Computer vision
5. Robotics

Differentiate between human intelligence and machine intelligence in terms of

1. Origin- Human intelligence is biological (i.e.; evolved from complex structure and processes of human brain) , while machine intelligence is artificial.(i.e.; created by humans through algorithms and programming)

2. Learning- Human intelligence learns through experience, intuition and social interaction , while Machine intelligence through data ,programming and algorithms

3. Awareness- Humans are conscious with feelings and self-awareness ,while machines do not have consciousness or emotions – they work on instructions

4. Creativity- Human intelligence possesses creativity , imagination and innovation , while machine intelligence has limited creativity within defined parameters.

5. Adaptability-Human intelligence quickly adapts to new situations, whereas machines need to be reprogrammed to handle new tasks.

6. Social skills- Humans understand social dynamics, emotions and can build relationships. Machines lack emotions and has limited social understanding.

7. Senses- Humans have various senses which help gather information, whereas machines have limited sensors.

8. Context- Humans understand the context of situations by considering the various factors , while machines require explicit programming to understand the context

9. Ethics and Morality- HI possess moral and ethical understanding. MI lacks moral and ethical understanding.

What is Turing Test?

The Turing test developed by Alan Turing in 1950, is a measure of a machines ability to develop human like intelligence.

The test assesses a machines ability to think intelligently like a human and is widely used as a bench mark for evaluating the progress of artificial intelligence research.

Role of Turing test in AI

Turing test is used to determine whether a machine can think like a human.

In the test a human judge will engage in a text based conversation with both a human and a machine, and then decide which respondent is human and which is machine. If the judge is unable to distinguish between human and machine based on the conversation, then the machine is said to have passed the Turing test.

Uses/ Applications of Turing test

Humans can be biased in judgments but, Turing test being conducted in controlled procedures and environment with hidden participants details, it proves to be unbiased and accurate.

Captcha is one type of turing test to find out whether the user is a computer or a human.

Importance of Turing test

It can be used to find out how advanced computer AI is nowadays.

Connectivity between human intelligence and machine intelligence

Human intelligence is the inspiration of machine intelligence. As humans continue to develop advance artificial intelligence technologies , machines with increased computational power and data processing abilities are created and thus the gap between human and machines are reduced .

Examples : Autonomous vehicle and Personal assistance

What is Cyber space?

It is an unreal, non physical ,virtual complex environment on the internet , where software ,hardware and people interact by means of technology through devices and network connections on it.

What is cyber security?

Cyber security is a combination of technologies , practices and processes designed to protect data from getting damaged or someone getting unauthorized access to .

Cyber security involves a range of technologies, processes and practices including:

Network and cloud security

Application security

Identity management

Database and infrastructure security

Endpoint security

Mobile security

Data recovery planning

Types of cyber threats-

1. Malware- Files or programs designed to harm a computer e.g.: Virus, Worms Trojan horses, Spyware etc.
2. Ransomware – Is a type of malware that encrypts a victim's file or locks their device and demands a ransom in exchange for the decryption key or unlock code.
3. Phishing- is done through fraudulent emails, texts or messages that appear to come from a legitimate source.

What is cyber crime?

It refers to any criminal activity that uses a computer ,device or the internet to commit or facilitate a crime

Mention some ways to prevent (protect from)cyber crime

Use authentic internet security and antivirus soft wares

Regularly update the software

Create strong encrypted passwords

Use social media settings to prevent unauthorized use of personal information

Proper education about cyber threats and its dangers

Knowledge about rights and options to fight cyber crimes

Keep watch on kids, as they can become easy victims

What is a computer virus?

Computer virus (Vital Information under Siege) is a malware which spreads between computers or networks and causes damage to hardware, software and data.

Security and Ethical Issues

Unauthorized access of computer systems devices or equipment is a security risk as it can lead to data breaches,privacy violations and disruption of services. It is illegal and can result in legal consequences. Ethically ,it is considered wrong as it violates privacy,consent and ownership principles. To prevent unauthorized use security measures like strong passwords, authentication methods and physical locks should be implemented.

Theft of Software

Stealing of software media, illegally registering/activating/copying a program, or purposefully erasing a program, or any such activity of using or working with software for which one does not have ownership or authority to use.

Disputed Rights to Products

Any piece of software /data which is copyrighted and is an Intellectual Property, cannot be used wholly or partly without the authorization or permission from the owner. Any unauthorized use of the same fall under the purview of disputed rights to the product.

The Use of Computer to Commit Fraud

Is the use of internet services and internet devices on a computer to defraud individuals or organizations of their resources, which could be financial, data, intellectual property or even causing physical harm?

Hacking and Data Theft

Hacking is the act to break the security code and enter a computer system or network in order to exploit the information available for illicit purpose

Data theft is one of the outcomes of hacking where confidential or personal data of the individuals or corporations are stolen for the purpose of misuse.

Give two examples for cyber sabotage

Ransomware and cyber attack